

Practical-3:

Task-1:

Compilation and Execution:

```
p_harini@DESKTOP-RJCS74B: ~  
p_harini@DESKTOP-RJCS74B:~$ nano practical3.c  
p_harini@DESKTOP-RJCS74B:~$ gcc practical3.c  
p_harini@DESKTOP-RJCS74B:~$ ./a.out  
Before fork()  
PID: 988  
PPID: 883  
Process State: Running  
  
Parent Process  
PID: 883  
Child PID: 989  
Process State : Running  
  
Child Process  
PID: 989  
PPID: 877  
Process State: Running  
p_harini@DESKTOP-RJCS74B:~$
```

Code:

```
p_harini@DESKTOP-RJCS74B: ~  
GNU nano 8.7.1 practical3.c  
#include <stdio.h>  
#include <unistd.h>  
int main()  
{  
    int pid;  
    printf("Before fork()\n");  
    printf("PID: %d\n", getpid());  
    printf("PPID: %d\n", getppid());  
    printf("Process State: Running\n");  
    pid=fork();  
    if(pid<0)  
    {  
        printf("Fork failed\n");  
    }  
    if(pid == 0)  
    {  
        printf("\nChild Process\n");  
        printf("PID: %d\n", getpid());  
        printf("PPID: %d\n", getppid());  
        printf("Process State: Running\n");  
    }  
    else  
    {  
        printf("\nParent Process\n");  
        printf("PID: %d\n", getpid());  
        printf("Child PID: %d\n", pid);  
    }  
}
```

Task-2:

ps command:

```
p_harini@DESKTOP-RJCS74B: ~  
p_harini@DESKTOP-RJCS74B:~$ ps  
  PID TTY          TIME CMD  
   883 pts/2        00:00:00 bash  
  1003 pts/2        00:00:00 ps  
p_harini@DESKTOP-RJCS74B:~$
```

top command:

```
p_harini@DESKTOP-RJCS74B: ~  
p_harini@DESKTOP-RJCS74B:~$  
p_harini@DESKTOP-RJCS74B:~$ top  
top - 11:06:13 up 23 min, 1 user, load average: 0.00, 0.01, 0.00  
Tasks: 24 total, 1 running, 23 sleeping, 0 stopped, 0 zombie  
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st  
MiB Mem : 7784.6 total, 7155.3 free, 543.0 used, 237.2 buff/cache  
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 7241.6 avail Mem  


| PID  | USER     | PR | NI | VIRT   | RES   | SHR   | S | %CPU | %MEM | TIME+   | COMMAND         |
|------|----------|----|----|--------|-------|-------|---|------|------|---------|-----------------|
| 1    | root     | 20 | 0  | 24556  | 15512 | 11588 | S | 0.0  | 0.2  | 0:01.44 | systemd         |
| 2    | root     | 20 | 0  | 3180   | 2208  | 2072  | S | 0.0  | 0.0  | 0:00.01 | init-systemd(Ub |
| 6    | root     | 20 | 0  | 3180   | 2048  | 1940  | S | 0.0  | 0.0  | 0:00.00 | init            |
| 52   | root     | 19 | -1 | 50400  | 16792 | 15524 | S | 0.0  | 0.2  | 0:00.36 | systemd-journal |
| 84   | systemd+ | 20 | 0  | 22416  | 14488 | 11988 | S | 0.0  | 0.2  | 0:00.17 | systemd-resolve |
| 91   | root     | 20 | 0  | 35364  | 12344 | 9256  | S | 0.0  | 0.2  | 0:02.27 | systemd-udev    |
| 113  | root     | 20 | 0  | 2888   | 1944  | 1820  | S | 0.0  | 0.0  | 0:00.07 | chronyd-starter |
| 114  | root     | 20 | 0  | 4472   | 3128  | 2884  | S | 0.0  | 0.0  | 0:00.02 | cron            |
| 115  | message+ | 20 | 0  | 8808   | 5348  | 4688  | S | 0.0  | 0.1  | 0:00.10 | dbus-daemon     |
| 123  | root     | 20 | 0  | 44096  | 29848 | 14708 | S | 0.0  | 0.4  | 0:00.47 | networkd-dispat |
| 163  | root     | 20 | 0  | 18420  | 9348  | 8116  | S | 0.0  | 0.1  | 0:00.25 | systemd-logind  |
| 180  | syslog   | 20 | 0  | 220548 | 5960  | 4648  | S | 0.0  | 0.1  | 0:00.15 | rsyslogd        |
| 212  | _chrony  | 20 | 0  | 20424  | 6336  | 5560  | S | 0.0  | 0.1  | 0:00.09 | chronyd         |
| 218  | _chrony  | 20 | 0  | 12096  | 2444  | 1800  | S | 0.0  | 0.0  | 0:00.00 | chronyd         |
| 288  | root     | 20 | 0  | 123460 | 32604 | 17996 | S | 0.0  | 0.4  | 0:00.37 | unattended-upgr |
| 291  | root     | 20 | 0  | 5492   | 2768  | 2580  | S | 0.0  | 0.0  | 0:00.01 | agetty          |
| 338  | root     | 20 | 0  | 8648   | 5620  | 4776  | S | 0.0  | 0.1  | 0:00.04 | login           |
| 385  | p_harini | 20 | 0  | 22748  | 13444 | 10988 | S | 0.0  | 0.2  | 0:00.20 | systemd         |
| 387  | p_harini | 20 | 0  | 22920  | 4092  | 2088  | S | 0.0  | 0.1  | 0:00.00 | (sd-pam)        |
| 416  | p_harini | 20 | 0  | 6136   | 5476  | 3928  | S | 0.0  | 0.1  | 0:00.05 | bash            |
| 876  | root     | 20 | 0  | 3184   | 1120  | 980   | S | 0.0  | 0.0  | 0:00.00 | SessionLeader   |
| 877  | root     | 20 | 0  | 3200   | 1256  | 1108  | S | 0.0  | 0.0  | 0:00.02 | Relay(883)      |
| 883  | p_harini | 20 | 0  | 6264   | 5536  | 3860  | S | 0.0  | 0.1  | 0:00.10 | bash            |
| 1025 | p_harini | 20 | 0  | 8304   | 6072  | 3896  | R | 0.0  | 0.1  | 0:00.00 | top             |


```

p_harini@DESKTOP-RJCS74B: ~

p_harini@DESKTOP-RJCS74B:~\$

p_harini@DESKTOP-RJCS74B:~\$ top

top - 11:06:40 up 24 min, 1 user, load average: 0.00, 0.00, 0.00

Tasks: 26 total, 1 running, 25 sleeping, 0 stopped, 0 zombie

%Cpu(s): 0.0 us, 0.1 sy, 0.0 ni, 99.9 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st

MiB Mem : 7784.6 total, 7151.1 free, 547.2 used, 237.2 buff/cache

MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 7237.4 avail Mem

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
91	root	20	0	35364	12344	9256	S	0.3	0.2	0:02.34	systemd-udevd
1	root	20	0	24556	15512	11588	S	0.0	0.2	0:01.44	systemd
2	root	20	0	3180	2208	2072	S	0.0	0.0	0:00.01	init-systemd(Ub
6	root	20	0	3180	2048	1940	S	0.0	0.0	0:00.00	init
52	root	19	-1	50400	16792	15524	S	0.0	0.2	0:00.36	systemd-journal
84	systemd+	20	0	22416	14488	11988	S	0.0	0.2	0:00.17	systemd-resolve
113	root	20	0	2888	1944	1820	S	0.0	0.0	0:00.07	chronyd-starter
114	root	20	0	4472	3128	2884	S	0.0	0.0	0:00.02	cron
115	message+	20	0	8808	5348	4688	S	0.0	0.1	0:00.10	dbus-daemon
123	root	20	0	44096	29848	14708	S	0.0	0.4	0:00.47	networkd-dispat
163	root	20	0	18420	9348	8116	S	0.0	0.1	0:00.25	systemd-logind
180	syslog	20	0	220548	5960	4648	S	0.0	0.1	0:00.15	rsyslogd
212	_chrony	20	0	20424	6336	5560	S	0.0	0.1	0:00.09	chronyd
218	_chrony	20	0	12096	2444	1800	S	0.0	0.0	0:00.00	chronyd
288	root	20	0	123460	32604	17996	S	0.0	0.4	0:00.37	unattended-upgr
291	root	20	0	5492	2768	2580	S	0.0	0.0	0:00.01	agetty
338	root	20	0	8648	5620	4776	S	0.0	0.1	0:00.04	login
385	p_harini	20	0	22748	13444	10988	S	0.0	0.2	0:00.20	systemd
387	p_harini	20	0	22920	4092	2088	S	0.0	0.1	0:00.00	(sd-pam)
416	p_harini	20	0	6136	5476	3928	S	0.0	0.1	0:00.05	bash
876	root	20	0	3184	1120	980	S	0.0	0.0	0:00.00	SessionLeader
877	root	20	0	3200	1256	1108	S	0.0	0.0	0:00.02	Relay(883)
883	p_harini	20	0	6264	5536	3860	S	0.0	0.1	0:00.10	bash
1025	p_harini	20	0	8304	6072	3896	R	0.0	0.1	0:00.02	top
1032	root	20	0	35368	6460	3336	S	0.0	0.1	0:00.00	(udev-worker)
1033	root	20	0	35368	6460	3336	S	0.0	0.1	0:00.00	(udev-worker)

/proc:

p_harini@DESKTOP-RJCS74B:~\$ /proc

-bash: /proc: Is a directory

p_harini@DESKTOP-RJCS74B:~\$